WORKING PAPER

Examining the heterogeneous link between decentralization and economic development

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Abstract

The relationship between decentralization and economic development has been widely studied, yet empirical findings remain inconclusive. This paper revisits the question by estimating an empirical growth model using a panel approach with grouped fixed-effects, which also allows us to endogenously identify latent clusters of countries that exhibit distinct economic responses to decentralization. The results reveal substantial heterogeneity that conventional economic and institutional variables do not explain. To investigate the sources of this heterogeneity, we complement the analysis with a Random Forest approach. The findings highlight underlying country characteristics, such as cultural diversity, urbanization, state capacity, transport infrastructure, economic complexity, and fiscal capacity, as key predictors of the observed cluster differences. Overall, our evidence indicates that part of the economic consequences of decentralization depends on conditions not captured by standard institutional proxies.

Keywords: Decentralization; economic development; grouped fixed effects; random forest analysis.

JEL codes: H77; O43; O47.

1 Introduction

Few questions in political economy have generated as much empirical research—and as little consensus—as the relationship between decentralization and economic performance. For every study documenting growth benefits from devolving authority to subnational governments, another finds the opposite or no effect at all.

One possibility is that decentralization does not have a uniform effect because it does not operate in isolation from the broader political-economy environment. Whether devolution enhances or undermines performance may depend on underlying context-specific configurations that shape how authority is exercised once devolved. Recent work has emphasized these deeper conditioning factors. Rodríguez-Pose (2024) argue that centrifugal pressures and territorial politics fundamentally shape decentralization outcomes in ways that standard empirical research has failed to capture. Treisman (2007) similarly highlights how the performance of decentralized systems hinges on context-specific factors—intergovernmental competition, information structures, coordination capacity—that vary substantially across countries and are difficult to generalize.

If these insights are correct, then the inconsistent empirical findings may reflect unobserved heterogeneity that conventional approaches fail to capture adequately. The typical panel regression imposes a single average effect, pooling countries where decentralization succeeds with those where it fails. Even specifications that interact decentralization with observable institutional variables may miss the relevant variation if the conditioning factors are multidimensional or operate through channels that standard proxies do not measure well (Glaeser et al., 2004; Voigt, 2012).

There is also a temporal dimension that deserves attention. The benefits or costs of decentralization need not be constant over time, i.e., what works at one stage of institutional development or under one set of political conditions may become less effective—or even counterproductive—as circumstances evolve. This dynamic aspect is central to political-economy reasoning: the question goes beyond whether to decentralize or not, but when

and under what conditions devolution is likely to enhance performance. [1]

These considerations call for an empirical approach that allows heterogeneity without imposing strong assumptions about its sources or functional form. In this paper, we use grouped fixed-effects models following Bonhomme and Manresa (2015) to let the data reveal distinct patterns in how decentralization relates to growth across countries and over time. Analyzing 80 countries between 1996 and 2018, we find evidence consistent with substantial heterogeneity: some countries benefit from decentralization in terms of growth whereas others do not. This pattern persists when we control for conventional institutional measures like rule of law and democracy; in fact, 86% of countries remain in the same group after adding these controls. This stability suggests that the heterogeneity we document may operate through channels not well captured by standard institutional proxies and is, in our view, linked to underlying political-economy forces that need to be taken into consideration.

What might explain these different patterns? Our exploratory Random Forest analysis points toward structural characteristics—cultural diversity, state capacity, urbanization, infrastructure, economic complexity—that appear relevant for understanding group membership. These findings are broadly consistent with the emphasis in recent political-economy work on deep, multidimensional conditioning factors that shape how decentralization reforms play out in practice. Our results suggest caution against universal prescriptions for decentralization. If the relationship with growth varies substantially across contexts in ways that observable institutions do not fully explain, then policy recommendations need to be carefully tailored to specific country circumstances.

The rest of the article is structured as follows. Section 2 discusses theoretical perspectives on why countries decentralize and why the growth effects might vary across contexts. Section 3 presents our empirical strategy. Section 4 describes the data and measurement. Section 5 reports the main results and Section 6 concludes with a discussion of implications.

[1] See Rodríguez-Pose (2024) for a discussion about static and dynamic drivers of decentralization.

2 Theoretical Background

2.1 Why Do Countries Decentralize?

Decentralization is often shaped by a range of underlying incentives, some rooted in normative arguments about efficiency and others emerging from strategic political considerations. These motives interact in complex ways, and their relative importance may differ across countries and over time.

a) Efficiency and informational gains

One view emphasizes the *economic* advantages of decentralized governance. When preferences and cost structures vary locally, delegating authority can improve allocative outcomes by aligning decisions with conditions on the ground (Oates, 1972), which can, in turn, enhance local government finances (Asatryan et al., 2015).

Mobility of residents and firms adds another dimension: in the spirit of Tiebout (1956), decentralization fosters interjurisdictional competition, potentially curbing rent extraction and encouraging policy innovation (Hayek, 1939). Greater fiscal autonomy may also constrain the expansionary tendencies of a central Leviathan (Weingast, 1995), specially in democratic settings (Qiao et al., 2019).

Yet these arguments rest on demanding assumptions—credible budget constraints, limited externalities, effective coordination across layers of government and politicians acting as social-welfare maximisers (Weingast, 2014). Without them, decentralisation can fragment authority, generate inefficiencies, and exacerbate common-pool problems (e.g., Prud'homme, 1995; Mitra and Chymis, 2022). Hence, the net growth effect of efficiency-motivated decentralization is theoretically positive but practical application may be challenging.

b) Political incentives and identity claims

Other accounts view decentralization as a strategic response to *political* pressures. When central elites face challenges to their authority—whether from coalition politics, internal dissent, or regional demands—delegating power can be a way to share rents, shift blame, or contain opposition (Enikolopov and Zhuravskaya, 2007; Flamand, 2019). In ethnically or linguistically diverse societies, demands for autonomy often reflect not just programmatic preferences but also group-based claims to self-rule (Hechter, 2000; Faguet, 2014; Belmonte et al., 2018). Institutional concessions may defuse tensions, but they can also embolden secessionist movements (Gibilisco, 2021) or facilitate elite capture at the local level, especially when solidarity mechanisms remain weak (Brancati, 2006; Aparicio-Pérez et al., 2025). Ultimately, the speed, depth, and shape of devolution vary widely, hinging on country-specific mixes of territorial identities, state-building legacies and the spatial distributions of wealth (see, e.g., Treisman 2007; Rodríguez-Pose 2024). Therefore, from a growth perspective, these political factors could make the sign of the economic impact of decentralization a priori ambiguous, depending on country-specific characteristics.

2.2 Does Decentralization Foster or Hinder Economic Performance?

Cross-country evidence does not yield a single, unambiguous conclusion regarding the impact of decentralization on economic development. Early empirical contributions, such as Davoodi and Zou (1998) and Rodríguez-Pose and Ezcurra (2011), associate decentralization with weaker economic performance. By contrast, other studies, including Iimi (2005), Buser (2011), Ligthart and Van Oudheusden (2017), and Canavire-Bacarreza et al. (2020), document positive economic effects. A third strand of the literature, however, finds no statistically significant effects, reports results that are not robust across specifications, or suggests that the relationship is nonlinear and/or contingent on intervening factors (e.g., Thornton, 2007, Asatryan et al., 2015, Canare, 2022).

Indeed, this lack of consensus is well acknowledged in meta-analytic and survey-based contributions. For instance, Baskaran et al. (2016) show that estimated effects of decentralization vary widely across studies and conclude that more precise empirical strategies are required. Similarly, the survey by Martinez-Vazquez et al. (2017) documents substan-

tial dispersion in reported results and highlights several potential explanations, including the difficulty of accurately measuring jurisdictional heterogeneity (e.g., Salmon, 2013) as well as the complexity and partial unobservability of many institutional arrangements (e.g., Voigt and Blume, 2012).

3 Empirical Strategy

3.1 Decentralization and economic development

As previously noted, cross-country studies have produced a wide range of findings regarding the impact of decentralization on economic development. Likewise, the motivations behind governments' decisions to devolve authority are highly heterogeneous and shaped by diverse, context-specific factors. We argue that these divergent outcomes may reflect the influence of unobserved and/or difficult-to-measure forces that vary across countries and evolve over time.[2] Once such heterogeneity is acknowledged, it becomes entirely plausible that decentralization may appear growth-enhancing in some contexts, detrimental in others, and even neutral in certain cases.

If this situation holds, then the standard fixed-effects specification—one of the workhorse methods in panel data analysis—would not be suitable for dealing with such heterogeneity. A one-way country (effects) model removes permanent cross-country differences but assumes they are frozen in time; the two-way extension adds common year shocks yet still imposes one decentralization slope for every country and year. If the true response varies, these models merely deliver a smoothed-out mean[3]. To let the data reveal richer heterogeneity without exploding the parameter count, we adopt the grouped fixed-effects estimator of Bonhomme and Manresa (2015). This approach is particularly useful for endogenously partitioning countries into latent clusters that share both a time-varying unobserved component and—when permitted—their own slope coefficient for decentralization. In practical terms, this enables countries whose residual growth trajectories

[2]Examples include state capacity, territorial or ethnic fragmentation, secessionist pressures, and the credibility of intergovernmental fiscal arrangements; our framework remains agnostic regarding their specific form.

[3]See Arkhangelsky and Imbens (2024) and Bellemare and Millimet (2025) among others, for discussion of treatment effects heterogeneity in fixed-effects frameworks. Their formal bias results are derived for binary treatments, but the intuition carries over to continuous regressors when marginal effects are suspected to be heterogeneous.

co-evolve—potentially due to shared unobserved factors such as increasing territorial fragmentation, infrastructure development, rising secessionist pressures, or synchronized institutional reform—to be grouped together. This framework thus allows heterogeneous economic responses to decentralization to emerge organically from the data.[4]

Formally, we estimate the following grouped fixed-effects model using the iterative algorithm of Bonhomme and Manresa (2015). The specification endogenously partitions countries into latent groups with distinct time-varying patterns of economic development and allows for group-specific decentralization effects (hereafter HGFE):

$$log(y_{it}) = x'_{it}\beta + \sum_{g=1}^{G} \gamma_g \cdot d_{it} \cdot \mathbf{1}\{g(i) = g\} + \alpha_{g(i)t} + \epsilon_{it}, \tag{1}$$

where $\log(y_{it})$ denotes the logarithm of real GDP per worker in country i and year t. The vector x_{it} includes standard growth and institutional controls, namely physical capital investment, human capital, the growth rate of the working-age population, and indices capturing the rule of law and democratic institutions. The variable d_{it} is the key regressor of interest and measures the degree of decentralization. The term $\alpha_{g(i)t}$ captures group-specific time-varying unobserved heterogeneity, with group assignments $g(i)$ endogenously estimated by the iterative algorithm, which clusters countries exhibiting similar income per capita dynamics net of observed covariates. The parameter γ_g identifies the decentralization effect specific to group g, thereby allowing the marginal impact of decentralization on economic development to vary across latent country clusters. Finally, ϵ_{it} is an idiosyncratic error term.

Previous work on the relationship between decentralization and economic development emphasizes that *institutions*, typically proxied by rule of law and/or democracy, can shape both the extent of decentralization and its economic impact (Enikolopov and Zhuravskaya, 2007; Buser, 2011). Motivated by the "hierarchy-of-institutions" hypothesis (Acemoglu et al., 2005; Flachaire et al., 2014), we consider two alternative model specifications: a restricted specification that omits institutional proxies for the rule of law and democracy, and the baseline model presented in Equation (1). If institutional qual-

[4]See, for instance, Aparicio-Pérez and Ripollés (2025); Gomez-Puig et al. (2022); Johar et al. (2022) for more examples on the use of this methodology.

ity were the primary source of heterogeneity in the decentralization–growth relationship, then including measures of rule of law and democracy should substantially alter the group assignments. Otherwise, we would expect the clustering structure to remain broadly stable.

3.2 Endogeneity

Endogeneity concerns may arise in Equation (1) due to simultaneity between economic development (y_{it}) and the following key covariates: decentralization (d_{it}), rule of law ($\log(strl_{it})$), and democracy ($\log(dem_{it})$). In the context of growth models, the two former are widely recognized as fundamental drivers of development, but they may also evolve in response to economic conditions, creating potential reverse causality (Acemoglu et al., 2001; Przeworski et al., 2000; Gerring et al., 2005). Decentralization, for example, may enhance growth by improving governance, accountability, and local service delivery (Faguet, 2014; Fisker, 2017). However, more developed countries or regions often have greater fiscal capacity, stronger state institutions, and higher political demand for decentralization, implying a two-way relationship. Similarly, although the rule of law and democratic institutions are frequently cited as essential preconditions for sustained economic development (Hall and Jones, 1999; Barro, 1996; La Porta et al., 1999), rising income levels may also stimulate institutional reforms and democratization.

Failing to account for simultaneity may result in biased and inconsistent estimates. To address this concern, we replicate the analysis by using two-stage least squares (2SLS), implementing two alternative identification strategies. On the one hand, we first instrument decentralization, rule of law, and democracy with their own lagged values, following Reed (2015). On the other hand, as an alternative, we use instrumental variables (IVs) based on exogenous predictors of institutional quality commonly used in the literature. Specifically, we rely on geographical factors and legal origins.

3.3 Exploring the potential drivers of group heterogeneity

After examining potential differences across country groups in the economic impact of decentralization, we turn to an analysis of the country-level *factors* that may be associated with the observed heterogeneity. At this stage, in the absence of a well-established theoretical framework, we adopt a data-driven exploratory approach based on a Random Forest classifier (Breiman, 2001). Specifically, we model the HGFE-based group assignment $G_i \in \{1, 2, 3\}$ of country i as a function of a set of candidate driving factors, denoted by $Z_i = (z_{i1}, \ldots, z_{ip})$.

The Random Forest is a flexible, non-parametric ensemble learning method that approximates complex classification functions by aggregating predictions from multiple decision trees grown on bootstrap samples and using randomly selected splits of the covariate space. By combining nonlinearities and high-order interactions among explanatory factors, the classifier assigns each country to a group by majority vote across trees. This framework enables us to jointly assess a broad set of socio-economic factors as potential correlates of group membership, while avoiding restrictive functional-form assumptions.

To investigate the contribution of individual factors to the classification of G_i, we rely on post-estimation interpretability tools. In particular, we compute global feature importance using SHAP values (Shapley Additive Explanations), obtained via Monte Carlo approximation following Greenwell (2022). These values quantify each factor's average marginal contribution to the model's predicted class probabilities. Finally, we employ Partial Dependence Plots (PDPs; Friedman, 2001) to illustrate how the predicted probability of belonging to each country group varies with changes in a given factor, holding all other factors at their observed values and averaging over the sample.

4 Data and Variables

For the main empirical analysis based on Equation (1), we construct a balanced panel of 80 countries covering the period 1996–2018, combining standard growth determinants with institutional and decentralization measures drawn from established sources. Firstly, from the Penn World Table (PWT version 10.01), we obtain information on real GDP per worker (y_{it}), private investment share of GDP (i_{it}), the human capital index (hc_{it}), and the annual growth rate of the working-age population. Following Mankiw et al. (1992), we augment the population growth rate by adding a fixed 5% term to capture depreciation and exogenous technological progress, yielding the effective population growth component (n_{it}).

Secondly, institutional quality is proxied by a standardized rule of law index ($strl_{it}$) and a liberal democracy index (dem_{it}), both sourced from the V-Dem Institute.[5] While the standardized rule of law index captures the extent to which legal rules are predictably enforced, the liberal democracy index measures the degree to which electoral democracy is constrained by institutional checks on political power.

Thirdly, decentralization (d_{it}) is measured using the *Regional Authority Index* (RAI) of Marks et al. (2008) and Hooghe et al. (2016). Based on ten institutional dimensions, the RAI captures both the "self-rule" of regions (e.g., institutional depth, fiscal autonomy) and their "shared-rule" influence over national policy (e.g., law-making, executive control). By jointly reflecting political, administrative, and fiscal authority (Ezcurra and Rodríguez-Pose, 2013), it provides a broad, theory-consistent measure of decentralization, in contrast to earlier studies focused narrowly on revenue or expenditure shares (see Jean-Baptiste Paul Harguindéguy and Pasquier, 2021). We use the country-year series from the public RAI_Country file (Shair-Rosenfield et al., 2021), which aggregates regional scores so the unit of analysis remains the country.[6] The RAI ranges from 0 (no regional authority) upward, with higher values indicating stronger regional authority. Table 1

[5] In the original V-Dem dataset, the rule of law index corresponds to v2x_rule and the liberal democracy index to v2x_libdem. For expositional simplicity, we denote these variables as $strl_{it}$ and dem_{it} throughout the paper.

[6] Version 3.1 of the dataset and codebook are available at https://garymarks.web.unc.edu/data/regional-authority-2/

reports descriptive statistics for the variables used in the main analysis. All variables in the baseline specification (Equation 1) enter the model in logarithms, except those that can take zero or negative values in our sample.

Table 1: List of data sources and descriptive statistics of panel variables used in the main analysis (Equation 1)

Source	Variable	Description	Mean	Std. Dev.	Min.	Max.
PWT	y_{it}	Real GDP per worker (constant US$).	49,360	34,572	2,364	198,368
PWT	i_{it}	Investment share of GDP (%).	23.688	6.573	0.449	60.328
PWT	hc_{it}	Human capital index.	2.783	0.583	1.354	4.154
PWT	n_{it}	Effective population growth rate (%).	6.484	2.188	-8.981	17.492
V-Dem	$strl_{it}$	Rule of law index (standardized).	0.639	0.187	0.041	1.000
V-Dem	dem_{it}	Liberal democracy index.	0.577	0.251	0.017	0.897
RAI	d_{it}	Regional Authority Index (decentralization).	10.856	9.839	0.000	37.720

PWT, V-Dem, and RAI refer, respectively, to the Penn World Table (version 10.01), the Varieties of Democracy (V-Dem) dataset, and the Regional Authority Index developed by Marks et al. (2008) and Hooghe et al. (2016).

Finally, in the second empirical analysis, we examine the determinants of the previously identified cross-country group heterogeneity using a cross-sectional dataset that includes a broad range of socio-economic factors for the same set of countries considered in the main panel analysis. Whereas the first analysis exploits the panel dimension of the data to identify groups of countries potentially exhibiting heterogeneous responses to decentralization, the second analysis takes these group assignments as given and seeks to account for them using cross-sectional information. Table 2 summarizes the factors considered in the second analysis and their data sources. Most variables are averaged over the period 1996–2018, subject to data availability, while others capture historical characteristics or pre-conditions in the sample.[7]

5 Results

5.1 Decentralization and economic development

Table 3 reports the estimates of the HGFE specification in Equation (1) and, for comparison, results from alternative panel-data models. We begin with a country fixed-effects (FE) specification and then extend it to a two-way fixed-effects model (TWFE) that

[7]The cross-sectional nature of this second analysis facilitates the inclusion of a wide set of contextual factors that are typically difficult to observe consistently over time. Even though, approximately 31% of the countries exhibit between one and three missing values, which are imputed using the corresponding cross-sectional mean. In any case, results from this second analysis should be interpreted as exploratory, as the objective is to assess which factors may be systematically associated, to a greater or lesser extent, with the heterogeneity in the decentralization–development relationship identified in the main empirical analysis.

Table 2: List of data sources and cross-sectional variables for the Random Forest classifier

Source	Variable	Description
Acemoglu et al. (2002)	`fmr_euro_colony`	Former European colony dummy.
Hausmann et al. (2014)	`eci`	Economic Complexity Index, average 1996–2018.
Fearon (2003)	`fe_etfra_pd`	Ethnic fractionalization: probability that two randomly selected individuals belong to different "ethnoreligious" groups ($\geq$ 1% of population), 1990s.
Fearon (2003)	`fe_cultdiv_pd`	Cultural diversity: modified `fe_etfra` to account for linguistic (cultural) distance between groups, 1990s.
Fearon (2003)	`fe_lmin_pd`	Share of the population belonging to the second-largest ethnic group (largest minority), 1990s.
Fearon (2003)	`fe_plural_pd`	Share of the population belonging to the largest ethnic group (plurality group), 1990s.
World Bank (2018)	`internet`	Share of the population using the internet, average 1996–2018.
Authors' coding	`island`	Dummy = 1 if the country is an island.
Putterman and Weil (2010)	`nonindigenous`	Dummy = 1 if more than 50% of the population is non-indigenous.
Hanson and Sigman (2021)	`state_capacity_estimate`	Composite index of state capacity to raise and manage revenue (extractive), control the use of force (coercive), and ensure bureaucratic effectiveness in policy and service delivery (administrative), average 1996–2015.
World Bank (2018)	`tax_revenue_perGDP`	Tax revenue as a percentage of GDP, average 1996–2018.
World Economic Forum	`transport`	Composite measure combining eight indicators covering the quality and extent of road, rail, air, and water transport infrastructure, average 2016–2018.
World Bank (2018)	`urbrate_1995`	Urban population as a percentage of the total population, year 1995.
Coppedge et al. (2024)	`v2xpe_exlgeo`	Exclusion by urban–rural location index (range 0–1), average 1996–2018.
Coppedge et al. (2024)	`v2xpe_exlpol`	Exclusion by political group index (range 0–1), average 1996–2018.

includes both country and time effects. We next implement the grouped fixed-effects estimator (GFE) of Bonhomme and Manresa (2015), which allows for time-varying group-specific unobserved heterogeneity at the country level. We report both the homogeneous-slope version (GFE) and its extension with heterogeneous decentralization slopes (HGFE). In both cases, we set the number of country groups to $g = 3$.[8] All specifications are estimated both without and with the inclusion of institutional quality controls, as described above.

In the standard FE and TWFE specifications (columns 1 and 2 of Table 3), decentralization exhibits a positive and statistically significant association with economic devel-

[8] As a robustness check, we have re-estimated Equation (1) using both the GFE and HGFE approaches under alternative group partitions, setting the number of latent groups to $g = 2$ and $g = 4$. These alternative results are not provided here for space reasons. Overall, country group assignments and the qualitative pattern of the estimated coefficients remain broadly consistent across specifications. Moreover, model comparisons based on the Bayesian Information Criterion (BIC) yield very similar values across different choices of g, indicating no meaningful improvement in fit from increasing or reducing the number of groups. We therefore adopt $g = 3$ in the baseline specification, as it provides a parsimonious and sufficiently rich partition that captures the three broad categories of possible economic effects of decentralization: positive, negative, or null.

opment. By contrast, in column 3, which reports estimates from the GFE specification allowing unobserved heterogeneity to evolve over time within endogenously identified country groups, the estimated coefficient on decentralization declines sharply and becomes statistically insignificant. This pronounced change suggests that the time-invariance assumption underlying the FE and TWFE models may be inappropriate in this context. Over a two-decade horizon, key country characteristics are unlikely to remain constant.[9] If such factors both affect economic development and correlate with decentralization while evolving over time, standard fixed-effects models will fail to account for them adequately. By allowing country-specific unobservables to follow group-specific time trajectories, the GFE framework provides a more flexible approach to capturing these dynamics.

[9]Secessionist pressures may intensify or dissipate over relatively short periods (Brancati, 2006); state capacity can strengthen or erode in response to political developments or fiscal crises (Hanson and Sigman, 2021); and economic structure can evolve substantially as countries move up the development ladder (Hausmann et al., 2014). Moreover, these processes often occur in waves: decentralization reforms themselves tend to be episodic (Rodríguez-Pose, 2024), frequently triggered by political transitions, economic shocks, or shifts in territorial identity (Flamand, 2019).

Table 3: Estimates of Equation (1)

	Without Institutional Proxies				With Institutional Proxies			
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
	FE	TWFE	GFE	HGFE	FE	TWFE	GFE	HGFE
$\log(i_{it})$	0.225***	0.182***	0.400***	0.409***	0.213***	0.168***	0.158***	0.176***
	(0.051)	(0.042)	(0.037)	(0.037)	(0.055)	(0.046)	(0.029)	(0.032)
$\log(hc_{it})$	2.811***	0.776***	2.915***	2.856***	2.856***	0.792***	2.131***	2.094***
	(0.088)	(0.078)	(0.034)	(0.033)	(0.102)	(0.113)	(0.069)	(0.064)
n_{it}	-0.002	-0.001	-0.014**	-0.014**	-0.005**	-0.003	-0.010**	-0.010**
	(0.003)	(0.002)	(0.006)	(0.006)	(0.002)	(0.002)	(0.004)	(0.004)
d_{it}	0.015***	0.011***	0.0006		0.011***	0.008***	0.001*	
	(0.003)	(0.002)	(0.0005)		(0.003)	(0.002)	(0.0007)	
$d_{it} \cdot \mathbf{1}\{g_i = 1\}$				0.006***				0.004***
				(0.001)				(0.001)
$d_{it} \cdot \mathbf{1}\{g_i = 2\}$				-0.008***				-0.007***
				(0.000)				(0.001)
$d_{it} \cdot \mathbf{1}\{g_i = 3\}$				-0.010***				0.001
				(0.002)				(0.001)
$\log(strl_{it})$					0.272***	0.241***	0.790***	0.798***
					(0.052)	(0.053)	(0.069)	(0.066)
$\log(dem_{it})$					-0.004	0.007	0.075***	0.070***
					(0.022)	(0.023)	(0.013)	(0.013)
Country FE	✓	✓			✓	✓		
Country and year FE		✓				✓		
Grouped FE			✓	✓			✓	✓
Observations	1886	1886	1886	1886	1600	1600	1600	1600
Countries	82	82	82	82	80	80	80	80
Adjusted R^2	0.561	0.064	0.681	0.687	0.579	0.097	0.815	0.817
F statistic	426.453	42.013	2152.536	6025.138	315.696	36.368	7086.155	8812.200
P-value	[0.000]	[0.000]	[0.000]	[0.000]	[0.000]	[0.000]	[0.000]	[0.000]

The dependent variables is the logarithm of real GDP per worker. Driscoll–Kraay robust standard errors are reported in parentheses, while p-values in brackets. FE, TWFE, GFE, and HGFE denote fixed effects, two-way fixed effects, grouped fixed effects, and grouped fixed effects with interactions between decentralization and group dummies, respectively.

However, the GFE specification still imposes the restriction that decentralization has a uniform effect across all countries. This assumption is at odds with the theoretical discussion in Section 2, which identified opposing channels through which decentralization can affect growth. If efficiency gains from better local information and competition dominate in some contexts, while coordination failures, elite capture, fictionalized societies or weakened state capacity dominate in others, then a single coefficient will obscure

these countervailing forces. To test whether such heterogeneity exists, we extend the GFE model to allow for group-specific decentralization slopes—the HGFE specification. The HGFE specification (column 4) reveals, indeed, substantial heterogeneity. The algorithm endogenously identifies three country groups with markedly different effects: one group where decentralization is associated with higher growth, another where the relationship is negative, and a third where it is even more adverse. This evidence seem to confirm that there are important group characteristics that drive opposing results in the decentralization-growth nexus. The composition of these groups are detailed in Table A1.

Columns 5 through 8 replicate the analysis including rule of law and democracy to control for *economic* and *political* institutions (Acemoglu et al., 2005). Interestingly, both variables enter with positive and highly significant coefficients in the HGFE specification (0.798 and 0.070, respectively), confirming their relevance in the process of economic development. However, the heterogeneity in decentralization effects not only persists but becomes more pronounced. Comparing column 4 to column 8, Group 1 retains a positive coefficient (falling from 0.006 to 0.004), Group 2 remains negative (-0.008 to -0.007), while Group 3's coefficient becomes statistically insignificant (from -0.010 to 0.001).

These results reveal substantial heterogeneity in the economic impact of decentralization that is largely masked by standard fixed-effects models. Comparing the FE/TWFE specifications (columns 1-2 and 5-6) with the grouped approaches (columns 3-4 and 7-8), we observe that imposing a single coefficient for all countries yields positive and significant effects, whereas allowing for time-varying group-specific heterogeneity uncovers stark differences: decentralization enhances income per capita in some country contexts, hinders it in others, and has no discernible effect in a third group. Critically, this pattern holds whether or not we control for institutional quality. The group composition remains largely stable (86% of countries retain their assignment, comparing Tables A1 and A2), indicating that the heterogeneity operates through channels not adequately captured by conventional measures of rule of law or democracy. This points toward underlying political-economy forces that shape how decentralization affects growth but are difficult to observe or measure directly.

To address potential endogeneity concerns, we re-estimate Equation (1) with HGFE using two instrumental-variable strategies, as outlined in Section 3. Table 4 reports the corresponding results. Column (1) presents the estimates obtained when the covariates d_{it}, $\log(strl_{it})$, and $\log(dem_{it})$ are instrumented with their own two lagged values, following Reed (2015). Column (2) shows the results when these variables are instrumented with a set of exogenous instruments, including geographical variables and legal origins indicators, from Nunn and Puga (2012). The geographical variables include terrain ruggedness (measured by the Terrain Ruggedness Index and the percentage of a country's land area that is highly rugged), latitude, and the percentage of each country's land surface with a tropical climate. The legal origins indicators include French civil law and Socialist civil law.[10] [11]

Legal origins provide historical determinants of contemporary institutional arrangements. Socialist legal origin predicts lower institutional quality (rule of law and democracy) due to the legacy of state control, weak protection of individual rights, and less effective mechanisms to control political activity (La Porta et al., 1999), as well as lower decentralization stemming from the tradition of centralized planning. French legal origin is particularly relevant for decentralization: the French civil law tradition developed around a highly centralized state model with professional judges controlled by the central government, in sharp contrast to the more decentralized common law system (Glaeser and Shleifer, 2002; Treisman, 2007). Both legal traditions were established through historical processes that predate our sample period, ensuring their exogeneity with respect to contemporary economic growth.

Geographical variables—terrain ruggedness, latitude, and tropical climate—are time-invariant physical characteristics that shaped the historical formation of states and institutional development but do not directly affect contemporary growth except through their influence on governance structures (Acemoglu et al., 2001; Nunn and Puga, 2012; Hall and Jones, 1999). Average terrain ruggedness at the country level, for instance,

[10] More information about the IVs available at https://diegopuga.org/data/rugged/

[11] The null hypothesis of underidentification is rejected, indicating that the model is identified. The weak identification statistic exceeds the critical threshold of 10, suggesting that instruments are relevant. The null hypothesis of the Hansen test is not rejected, supporting the validity and exogeneity of the instruments.

historically hindered both state centralization and the establishment of strong institutions (Nunn and Puga, 2012). Tropical climates and latitude are associated with colonial settlement patterns that determined whether extractive or inclusive institutions emerged (Acemoglu et al., 2001). Distance from the equator has also been linked to differences in institutional quality, with areas further from the equator developing stronger mechanisms of cooperation and social capital (Bjørnskov and Méon, 2015). These geographical factors jointly predict our three endogenous variables—decentralization, rule of law, and democracy—while remaining exogenous due to their physical, time-invariant nature.

Table 4: 2SLS estimates of Equation (1) with HGFE

	(1)	(2)
$\log(i_{it})$	0.217***	0.064
	(0.025)	(0.047)
$\log(hc_{it})$	2.143***	-0.480*
	(0.067)	(0.249)
n_{it}	-0.008*	0.005
	(0.004)	(0.010)
$d_{it} \cdot \mathbf{1}\{g_i = 1\}$	0.005***	0.030***
	(0.001)	(0.006)
$d_{it} \cdot \mathbf{1}\{g_i = 2\}$	-0.007***	-0.074***
	(0.002)	(0.018)
$d_{it} \cdot \mathbf{1}\{g_i = 3\}$	0.001	-0.007
	(0.000)	(0.011)
$\log(strl_{it})$	0.788***	2.649***
	(0.064)	(0.183)
$\log(dem_{it})$	0.063***	0.189*
	(0.017)	(0.091)
Grouped FE	✓	✓
Observations	1200	1200
Countries	80	80
Adjusted R^2	0.811	0.339
Kleibergen-Paap rk LM statistic	14.815	14.286
	[0.027]	[0.001]
Kleibergen-Paap rk Wald F statistic	385.924	45.630
Hansen J statistic	2.212	0.184
	[0.819]	[0.668]

The dependent variable is the logarithm of real GDP per worker. Estimates of Equation (1) with HGFE are reported under two empirical strategies. In column (I), the covariates d_{it}, $log(strl_{it})$, and $log(dem_{it})$ are instrumented with their own two lagged values, following Reed, 2015. In column (II), they are instrumented with exogenous instruments, including ruggedness (as index and % of high rugged terrain), latitud, tropical climate and legal origin indicators (french and socialist). Driscoll–Kraay standard errors are shown in parentheses, and p-values are reported in brackets. * $p < 0.10$; ** $p < 0.05$; *** $p < 0.001$.

The 2SLS results in Table 4 largely confirm our earlier findings. Across both instrumental-variables strategies, decentralization remains development-enhancing for Group 1 and detrimental for Group 2, while the estimated effect for Group 3 continues to be statistically insignificant. Measures of institutional quality (i.e., rule of law and democracy) also remain strongly and positively associated with development under both empirical strategies, highlighting their importance. Consistent with our previous results, the persistence of these group-specific patterns suggests that observable institutions alone do not explain

why decentralization promotes economic development in some contexts but hinders it in others, pointing instead to the role of deeper background conditions in driving this heterogeneity.

5.2 Exploring group heterogeneity

Building on the country groups identified by the HGFE estimator in Eq. (1),[12] we next employ a Random Forest classifier to examine which socio-economic characteristics listed in Table 2 are most informative for distinguishing across these group assignments. This last procedure does not aim to uncover causal mechanisms; rather, it provides an exploratory assessment of the variables that could contribute most to predicting the previously identified groups. The socio-economic predictors considered here serve as observable proxies for the two theoretical channels discussed in Section 2: efficiency-based factors (state capacity, infrastructure, urbanization, economic complexity, and fiscal resources) and identity-based political considerations (ethnic and cultural fractionalization). These variables inevitably remain imperfect indicators of deeper political-economy forces that are difficult to observe directly, including territorial tensions, intergovernmental coordination, the credibility of fiscal arrangements, perceived governmental legitimacy, informal power structures, and the timing of reforms. If such underlying forces were measurable with adequate precision, they would be incorporated directly into the main empirical specification. In their absence, the HGFE estimator indirectly captures their influence by identifying groups of countries that exhibit systematically different responses to decentralization, while the Random Forest allows us to identify which observable proxies could be most strongly associated with this grouping. Taken together, this approach offers suggestive evidence on the structural conditions that may underpin the identified heterogeneity in the decentralization–development nexus.

Figure 1 reports the relative importance of the 15 socio-economic predictors listed in Table 2 in discriminating between the three decentralization–development nexus groups. Predictor importance corresponds to each variable's percentage contribution to the total SHAP values, normalized to 100%. These results are obtained using the Random Forest approach with 2,000 trees, eight randomly selected predictors per split, and a minimum

[12] The complete list of country groups is reported in Table A2. These are the country groups according to the GFE estimates from Eq. (1) with institutional proxies.

terminal node size of five observations.[13]

The ranking indicates that a small subset of predictors accounts for a large share of the model's SHAP-based importance. Cultural diversity (`fe_cultdiv_pd`) receives the highest weight, contributing approximately one quarter of the total SHAP mass. It is followed by initial urbanization (`urbrate_1995`), the estimated state capacity index (`state_capacity_estimate`), transport infrastructure quality (`transport`), economic complexity (`eci`), and tax revenue as a share of GDP (`tax_revenue_perGDP`). In the classifier, these variables are those most heavily relied upon to separate observations across the three decentralization–development groups, reflecting that long-run structural and institutional features are particularly informative for the model's partition of the data.

A second tier of predictors contributes more modestly to the classification. Internet penetration (`internet`) and the population share of the largest minority (`fe_lmin_pd`), together with the political and geographic exclusion indices (`v2xpe_exlpol`, `v2xpe_exlgeo`) and the plurality index (`fe_plural_pd`), display non-negligible SHAP weights, but their contribution to the model's decision structure is clearly smaller than that of the leading variables.

The remaining predictors exhibit limited relevance for the classifier's partitioning of countries. Ethnic fractionalization (`fe_etfra_pd`) receives a relatively small share of total SHAP importance, while colonial legacy (`fmr_euro_colony`), insularity (`island`), and the non-indigenous population indicator (`nonindigenous`) play only a minor role in the model's classification. Overall, the SHAP distribution indicates that the Random Forest primarily relies on deep socio-economic and institutional features to distinguish between

[13] These hyperparameters were selected via Leave-One-Out Cross-Validation (LOOCV), which evaluates predictive performance by holding out each observation once as an independent test case. To address class imbalance, the training data were upsampled so that each group contributed an equal number of observations. Under the selected configuration, the model achieved the highest out-of-sample accuracy (65.7%), with a Cohen's Kappa of 0.50 and a balanced accuracy of 0.74. While these metrics reflect satisfactory performance given the limited sample and multiclass structure, they also suggest that additional, unobserved or hard-to-measure factors likely influence group differentiation—consistent with the presence of country-specific forces that vary across space and evolve over time. This further supports the use of the HDFE estimator in the main empirical analysis, as it effectively absorbs such unobserved heterogeneity.

decentralization–development groups, whereas purely geographic or historical markers contribute little to the model's assignment of observations to groups.

Figure 1: Relative importance of socio-economic factors for classifying the country groups

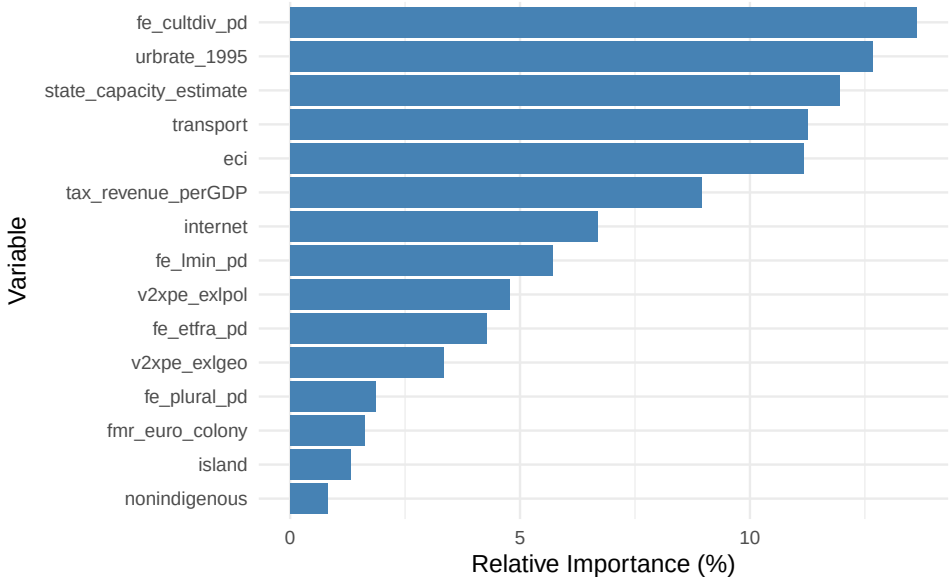


Figure computed from a Random Forest with 2000 trees using mean absolute SHAP values (normalized to 100%).

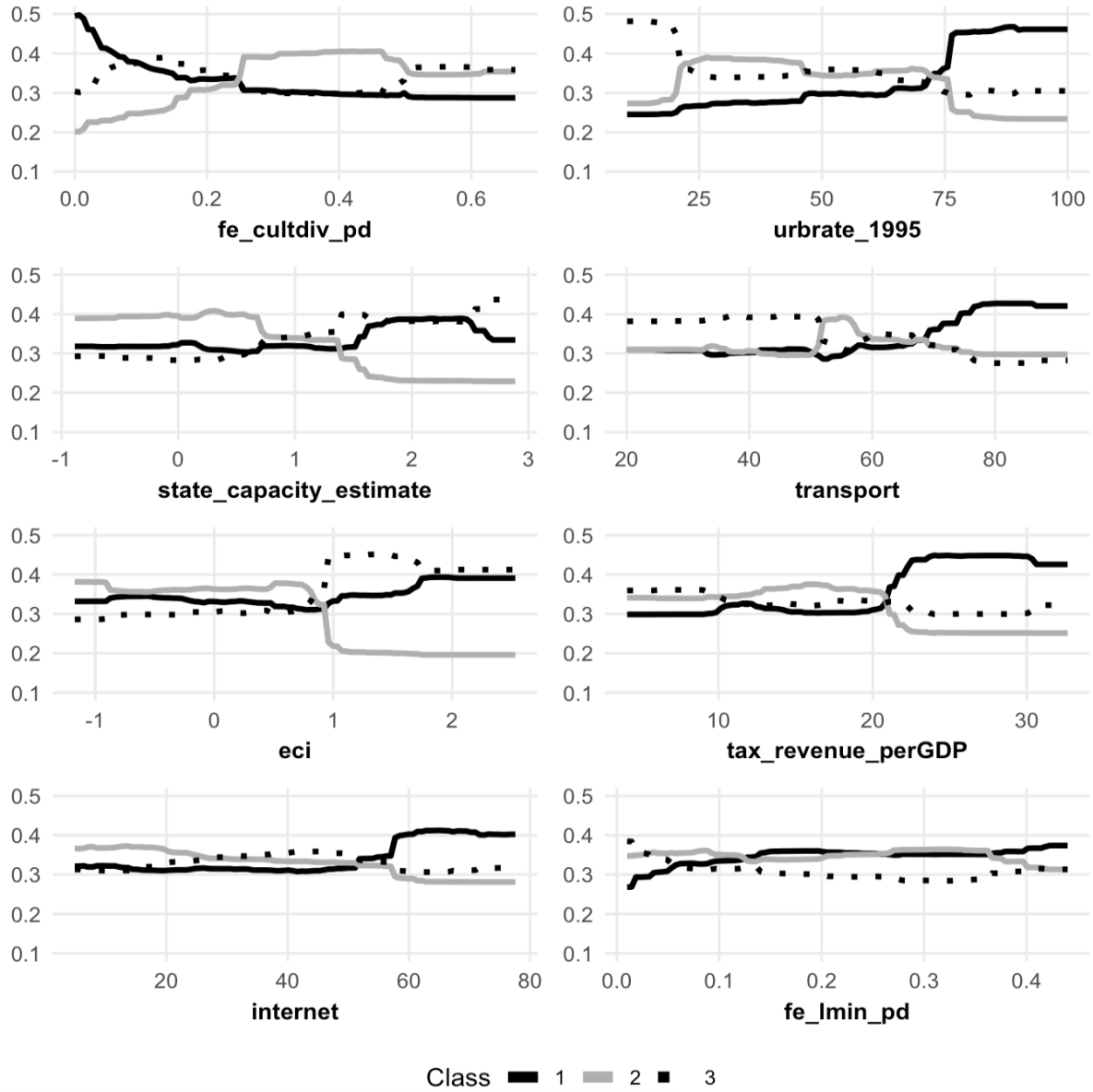
Finally, to further illustrate how the most relevant predictors contribute to the model's classification, Figure 2 presents PDPs for the eight variables with the highest SHAP importance. These plots display how the average predicted probability of belonging to each decentralization–growth group changes as a given predictor varies, holding all other variables at their observed values.

The PDP for `fe_cultdiv_pd` shows that higher levels of cultural diversity reduce the probability of classification into Group 1 (i.e., countries where decentralization enhances economic development) while increasing the probability of belonging to Group 2 (where decentralization is associated with lower levels of economic development). This pattern suggests that more heterogeneous societies may face greater coordination constraints or collective-action frictions that weaken (or even reverse) the potential developmental benefits of decentralization.

This pattern contrasts with that observed in the PDPs for urbanization (`urbrate_1995`), state capacity (`state_capacity_estimate`), transport infrastructure (`transport`), pro-

ductive complexity (`eci`), fiscal capacity (`tax_revenue_perGDP`), and digital penetration (`internet`), which indicate that higher values of these variables increase the likelihood of classification into Group 1 while reducing the probability of belonging to Group 2. Taken together, these latter results suggest that decentralization is more likely to generate stronger development outcomes in institutional and economic environments where subnational governments have greater capacity to manage resources, deliver public goods effectively, and coordinate investment.

Figure 2: Estimated effects of predictors on the average probability of belonging to each country group, based on the Random Forest classifier



Partial dependence plots (PDPs) from the Random Forest classifier, based on 2000 trees.

Beyond these results, the PDP for `fe_lmin_pd`, which has relatively low SHAP importance, indicates that ethnic fractionalization has little systematic association with group membership, as predicted probabilities remain largely flat across its range. Finally, the probability of belonging to Group 3 remains broadly stable across most predictor values, with only minor deviations. This stability is in line with the generally insignificant economic effects of decentralization observed for this group.

6 Conclusions

This paper has revisited the decentralization–development nexus by allowing for heterogeneous effects across countries. Using a panel of 80 countries over the period 1996–2018, we have estimated an augmented Solow growth model employing the grouped fixed-effects estimator proposed by Bonhomme and Manresa (2015). This approach has enabled us to identify latent clusters of countries characterized by distinct development trajectories and systematically different economic responses to decentralization. We have then applied a Random Forest classifier to explore the potential drivers of the observed heterogeneity across groups. Our analysis yields three main findings.

First, the relationship between decentralization and economic development is not uniform across countries. Our empirical results indicate that, for a subset of countries, decentralization is associated with higher economic development, whereas for another subset the association is negative. For a third group of countries, no statistically significant relationship is found.

Second, heterogeneity in economic responses to decentralization persists after controlling for conventional institutional proxies, such as rule of law and democracy. Indeed, 86% of the countries in our sample retain their estimated group assignment once these controls are incorporated, suggesting that the underlying sources of heterogeneity operate through channels not fully captured by standard institutional measures. These findings remain robust after accounting for potential endogeneity.

Third, our exploratory Random Forest analysis highlights a set of structural country characteristics, such as cultural diversity, urbanization, state capacity, transport infras-

tructure, economic complexity, and fiscal capacity, as key predictors of group membership. These factors, acting as proxies for deeper political economy conditions, appear to shape the environments in which decentralization enhances or undermines economic performance.

6.1 Policy Implications

The substantial heterogeneity we document suggests that universal prescriptions regarding decentralization are unlikely to be appropriate.

Context matters. The same institutional reform (devolving fiscal authority, transferring administrative functions, or granting political autonomy) may yield divergent outcomes depending on background conditions. Policymakers and international organizations should resist the temptation to transplant successful decentralization models from one context to another without careful assessment of local circumstances.

Sequencing and complementary reforms. Our results suggest that countries with lower state capacity or weaker infrastructure may benefit from strengthening these foundations before—or alongside—decentralization reforms. Devolving authority to subnational governments that lack the administrative capacity to manage resources effectively may prove counterproductive.

The politics of decentralization. The prominent role of cultural diversity in our analysis, combined with the persistence of heterogeneity after controlling for institutional quality, suggests that the *motivation* behind decentralization matters. When devolution is driven primarily by pressures to accommodate regional elites or territorial demands rather than by efficiency considerations, the growth effects may be weaker or even negative. Decentralization as a political concession to subnational interests, aimed at defusing secessionist pressures, sharing rents with regional power brokers, or responding to identity-based claims, does not automatically translate into improved economic performance. It may instead fragment authority and facilitate local elite capture without generating offsetting efficiency gains. This distinction between efficiency-driven and politically-driven decentralization, while difficult to operationalize empirically, appears central to understanding

the heterogeneous outcomes we document.

6.2 Directions for Future Research

Our findings open several avenues for future investigation. First, understanding the mechanisms through which background conditions shape decentralization outcomes requires more detailed case-study evidence. Why exactly does cultural diversity appear to undermine the growth benefits of decentralization? What specific coordination failures or political dynamics are at work? Comparative case studies of countries within our identified groups—examining pairs that share similar observable characteristics but responded differently to decentralization—could illuminate the causal pathways that our quantitative analysis cannot fully identify.

Second, disaggregating to the regional level within countries could reveal whether the heterogeneity we document at the cross-country level also operates within nations. Do some regions benefit from decentralization while others do not? What regional characteristics predict differential responses? This line of inquiry seems particularly promising for several reasons. It would allow researchers to hold constant many country-level institutional features while exploiting variation in regional characteristics. It would speak directly to policy questions about how to design intergovernmental fiscal arrangements when regions within the same country have different capacities or face different constraints. And it would provide a more granular test of the mechanisms we have identified—whether state capacity, infrastructure, or diversity operate similarly at the subnational scale. Extensions of the grouped fixed-effects methodology to regional panels could prove fruitful in this regard.

Finally, examining outcomes beyond GDP per capita—inequality, public service delivery, political stability, environmental performance—would provide a more complete picture of how decentralization affects development. In sum, the question "Does decentralization promote economic development?" may be less useful than "Under what conditions does decentralization promote economic development?" Our analysis suggests that the answer depends on deeper structural characteristics that shape how devolved authority is

exercised in practice. Both advocates and critics of decentralization may be correct, but in different settings. Recognizing this complexity is a necessary first step toward more nuanced and ultimately more effective development policy.

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Declaration of Competing Interests

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Declaration of generative AI and AI-assisted technologies in the writing process

During the preparation of this work the author(s) used ChatGPT and Claude in order to language-edit the text. After using this tool, the author(s) reviewed and edited the content as needed and takes full responsibility for the content of the publication.

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Table A1: List of countries by group, according to the GFE estimates from Eq. (1) without institutional proxies

Group 1	**Group 2**	**Group 3**
Argentina	Albania	Bangladesh
Australia	China	Bolivia
Austria	El Salvador	Brazil
Barbados	Estonia	Bulgaria
Belgium	Guyana	Cambodia
Belize	India	Chile
Brunei Darussalam	Lao People's DR	Colombia
Canada	Latvia	Croatia
Costa Rica	Lithuania	Denmark
Cyprus	Mongolia	Dominican Republic
Czech Republic	Myanmar	Ecuador
Finland	Romania	Hungary
France	Russian Federation	Indonesia
Germany	Trinidad and Tobago	Ireland
Greece	Ukraine	Malaysia
Guatemala	Vietnam	Malta
Haiti		Nepal
Honduras		Norway
Iceland		Panama
Israel		Paraguay
Italy		Peru
Jamaica		Philippines
Japan		Poland
Luxembourg		Singapore
Mexico		Slovakia
Netherlands		Sri Lanka
New Zealand		Thailand
Nicaragua		Turkey
Pakistan		Uruguay
Portugal		
Republic of Korea		
Slovenia		
Spain		
Sweden		
Switzerland		
United Kingdom		
United States		

Table A2: List of countries by group, according to the GFE estimates from Eq. (1) with institutional proxies

Group 1	**Group 2**	**Group 3**
Argentina	Albania	Bangladesh
Australia	Bolivia	Bulgaria
Austria	China	Cambodia
Barbados	El Salvador	Chile
Belgium	Estonia	Costa Rica
Brazil	Guyana	Croatia
Canada	India	Czech Republic
Colombia	Lao People's DR	Denmark
Cyprus	Latvia	Ecuador
Dominican Republic	Lithuania	Hungary
Finland	Mongolia	Indonesia
France	Myanmar	Ireland
Germany	Romania	Malaysia
Greece	Russian Federation	Malta
Guatemala	Sri Lanka	Nepal
Haiti	Trinidad and Tobago	Norway
Honduras	Ukraine	Panama
Iceland		Paraguay
Israel		Peru
Italy		Philippines
Jamaica		Poland
Japan		Slovakia
Luxembourg		Slovenia
Mexico		Switzerland
Netherlands		Thailand
New Zealand		Turkey
Nicaragua		Uruguay
Pakistan		Vietnam
Portugal		
Republic of Korea		
Singapore		
Spain		
Sweden		
United Kingdom		
United States		